

① Design Message Platform Application like WhatsApp.

1. Functional Requirement :-

1. Should be able to send ^{and receive} message 1:1 to any other person.
2. Can send ^{and receive} messages in groups (where more than one people is present).

3. Last Seen

4. you can send images / memes

5. Messages has been Read or received.

2. Non Functional Requirement :-

1. Scalability (✓)

Eventual
2. Consistency

3. Availability (✓)

4. Reliability or Fault Tolerant (✓)

5. Low Latency.

[3.] Capacity Estimation :-

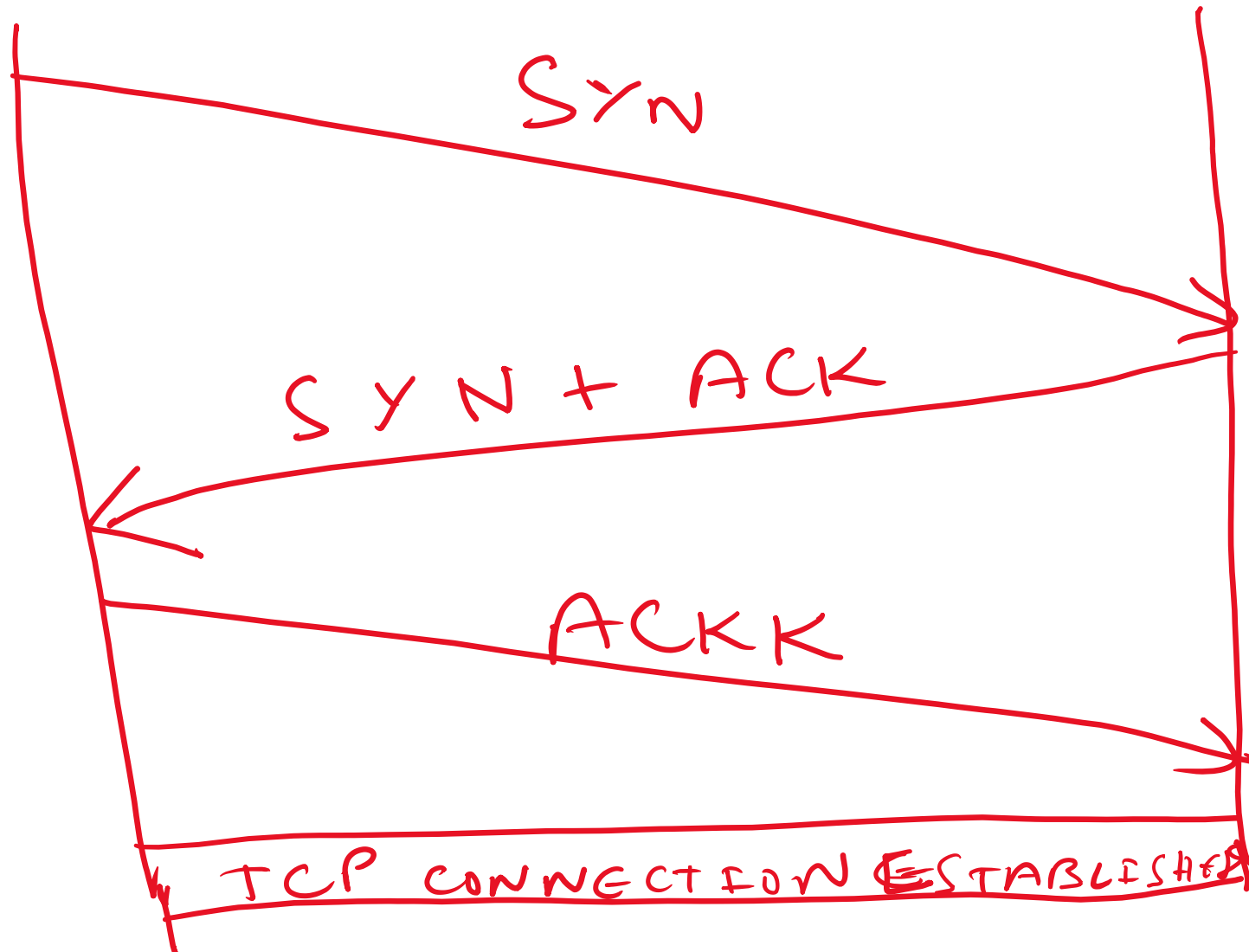
1. DAU :- In millions or in Billions.

2 - MAU :- In millions or in Billions.

①

Client

Server

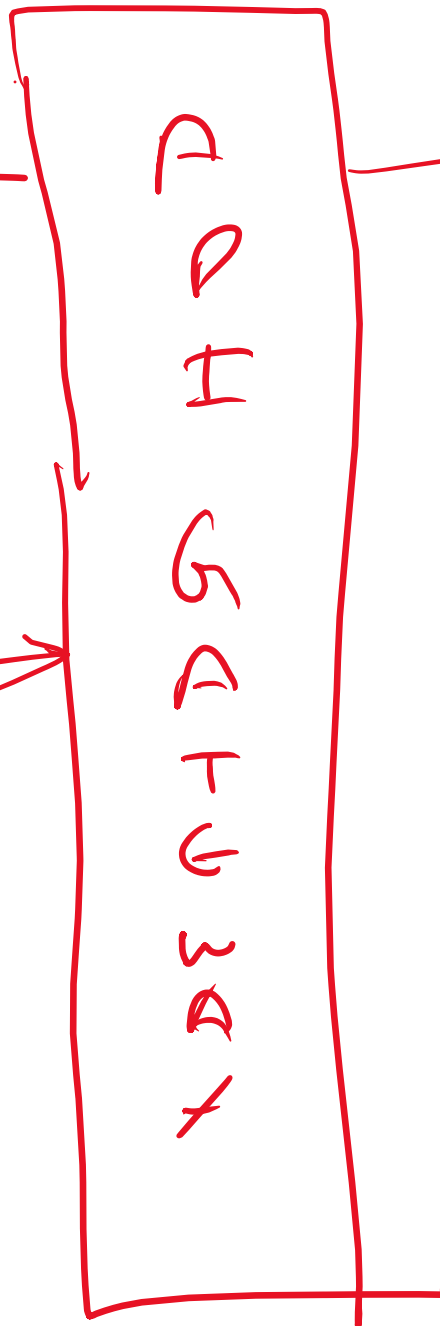
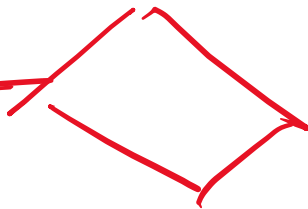


3-way
handshake

client 1

c1: c2: mm

client 2

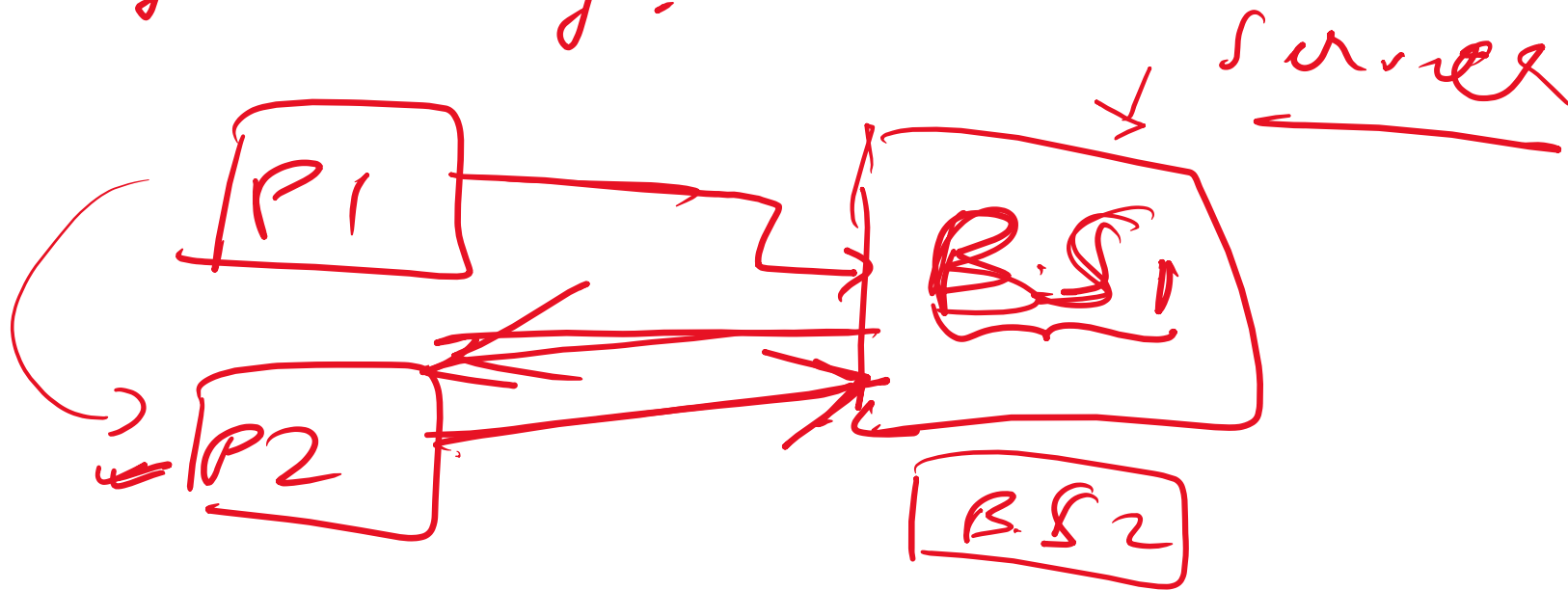


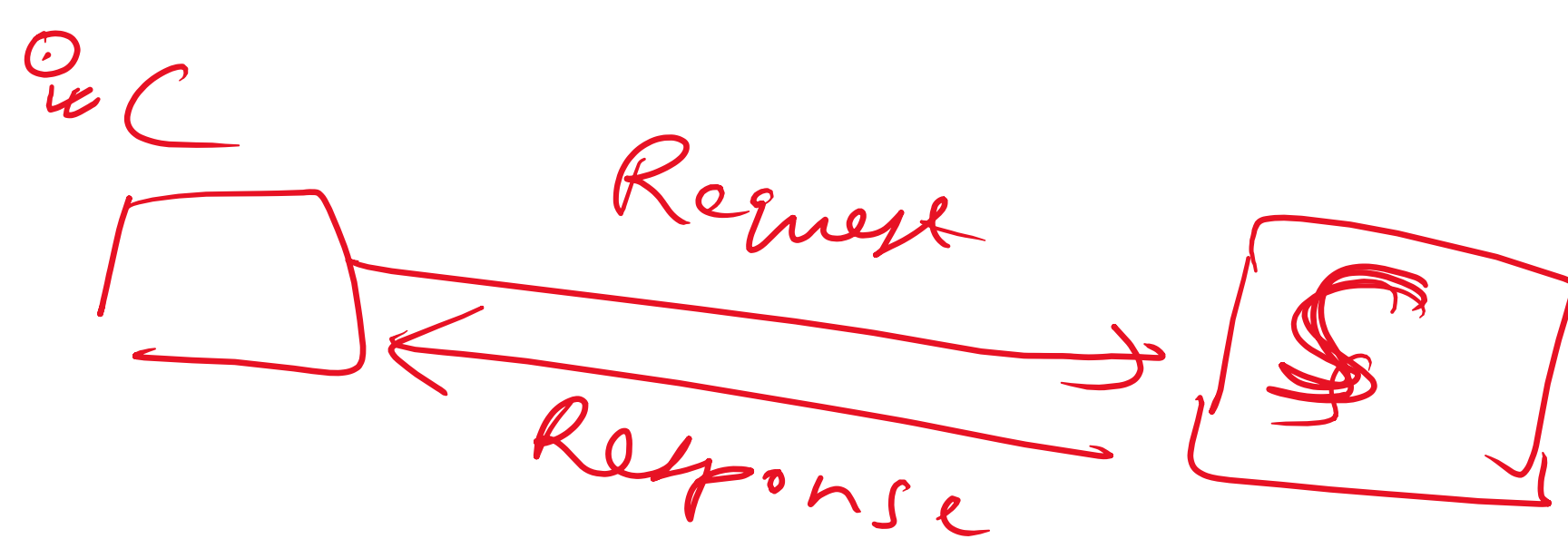
B.S

① The issue is that TCP connection needs to be established again & again.

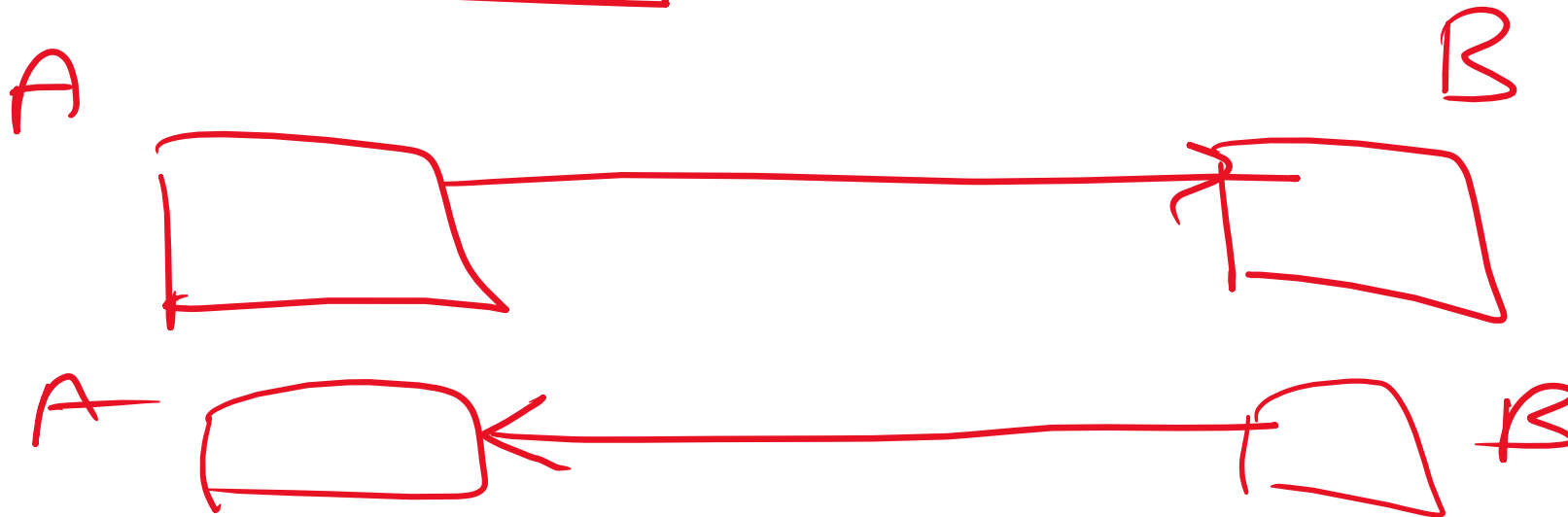
Solution :-

1. Long Polling :-

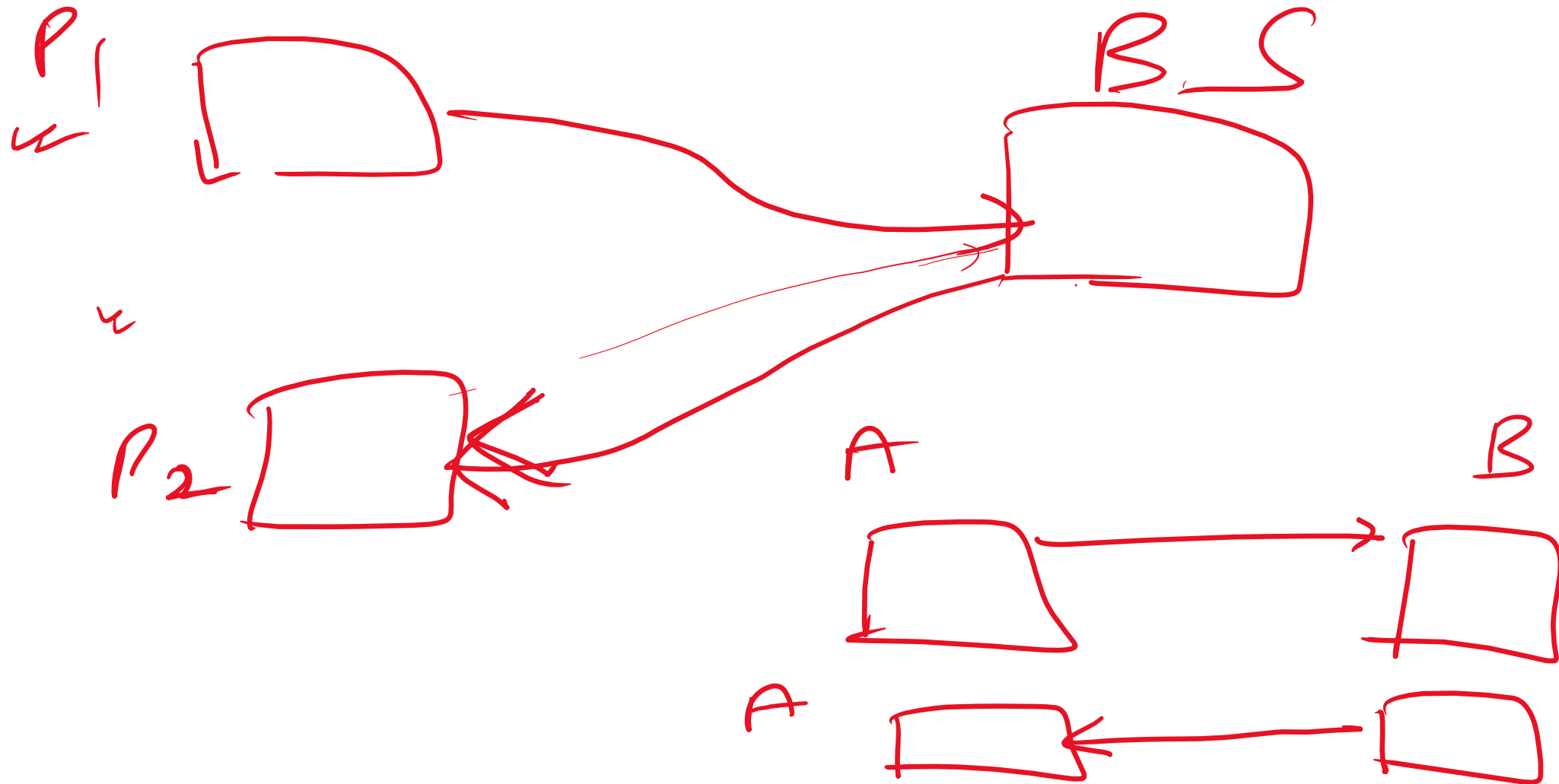


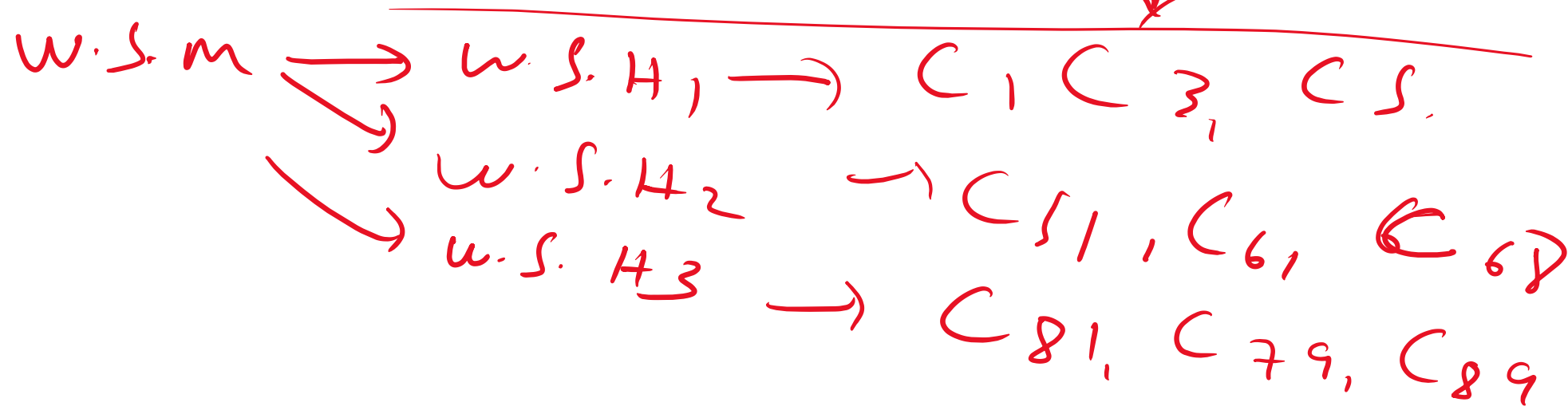
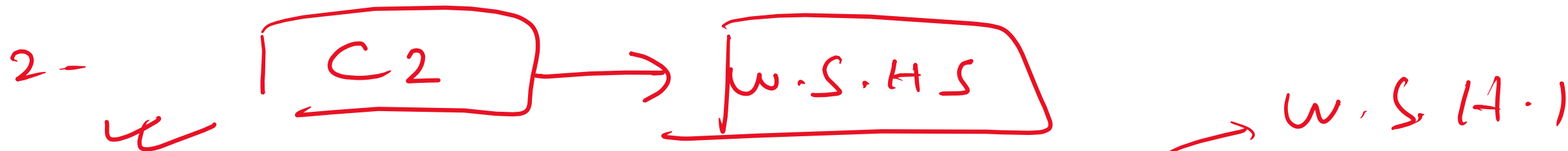


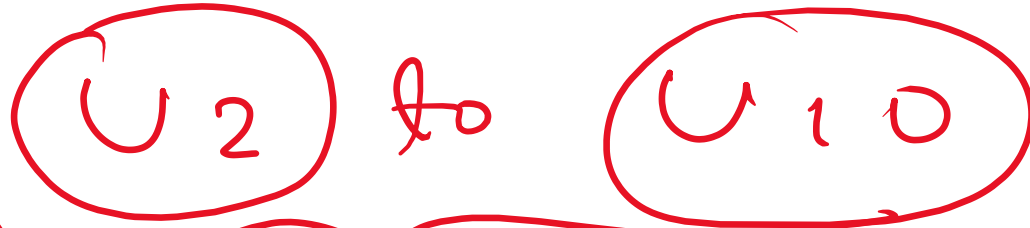
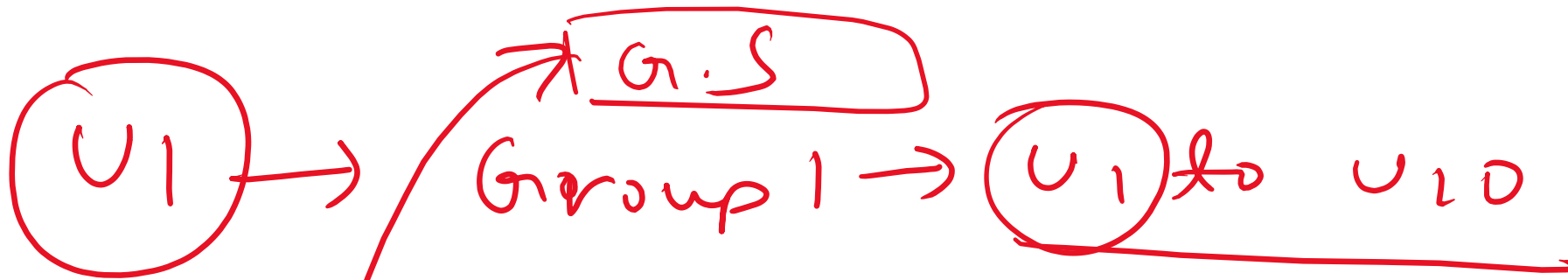
2. web Socket :-



○ Full Duplex Connection.







why websocket handler?



1 minute

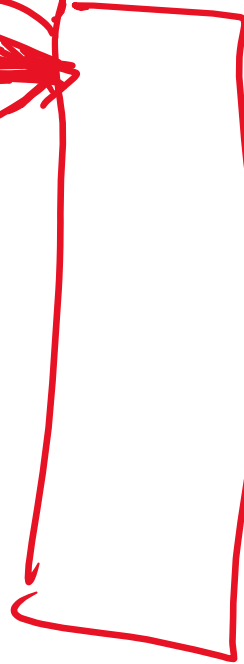
U.S.H.I

U1

flag=1

flag=0

U2



L.S.S

used, Timestamp
U1

11:30pm

30 sec